

EDEXCEL INTERNATIONAL A LEVEL

WME01/01 January 2021 Worked Solutions

January 2021 paper rebuilt with the worked solution after each question.

8

paper questions

WME01

Mechanics 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**M1**

PAST PAPER

WME01/01 January 2021

January 2021 | 8 questions | 75 marks

8

questions

75

marks

Answers

worked solution
after each
question

Question 1

Constant Acceleration in 1D

1. A small stone is projected vertically upwards with speed 20 ms^{-1} from a point O which is 5 m above horizontal ground. The stone is modelled as a particle moving freely under gravity.

Find

- (a) the speed of the stone at the instant when it is 2 m above the ground, (2)
- (b) the total time between the instant when the stone is projected from O and the instant when it first strikes the ground. (4)

Worked Solution - Question 1

Topic group

1. Find displacement for part a

The stone starts 5 m above the ground. When it is 2 m above the ground, its displacement from O is -3 m.

2. Use SUVAT for speed

Taking upward as positive, $u = 20$, $a = -9.8$ and $s = -3$. Use $v^2 = u^2 + 2as$:
 $v^2 = 20^2 + 2(-9.8)(-3) = 458.8$.

3. State the speed

$v = 21.4 \text{ m s}^{-1}$ to 3 significant figures.

4. Set up the ground equation

At the ground, displacement from O is -5 m. Use $s = ut + \frac{1}{2}at^2$:
 $-5 = 20t - 4.9t^2$.

5. Solve for time

$4.9t^2 - 20t - 5 = 0$. The positive root is $t = 4.3179\dots$, so the total time is 4.32 s.

Final answer

(a) 21.4 m s^{-1} . (b) 4.32 s approximately.

Question 2

Momentum, Impulse & Collisions

2. Two particles, P and Q , have masses $2m$ and m respectively. The particles are moving towards each other in opposite directions along the same straight line on a smooth horizontal plane. The particles collide directly.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $2u$.

The magnitude of the impulse exerted on Q by P in the collision is $5mu$.

Find

- (a) the speed of P immediately after the collision, (3)
- (b) the speed of Q immediately after the collision. (3)

Worked Solution - Question 2

Topic group

1. Choose a positive direction

Take P's original direction as positive. Before collision, P has velocity $3u$ and Q has velocity $-2u$.

2. Use impulse on P

The impulse on P is opposite to P's original motion and has magnitude $5mu$. Let P's velocity after collision be v . Then $-5mu = 2m(v - 3u)$.

3. Find P speed

Divide by m : $-5u = 2v - 6u$, so $2v = u$ and $v = \frac{u}{2}$.

4. Use impulse on Q

The impulse on Q is in P's original direction. Let Q's velocity after collision be w . Then $5mu = m(w - (-2u))$.

5. Find Q speed

$5u = w + 2u$, so $w = 3u$. Therefore the speed of Q is $3u$.

Final answer

(a) speed of P = $\frac{u}{2}$. (b) speed of Q = $3u$.

Question 3

Forces

3.

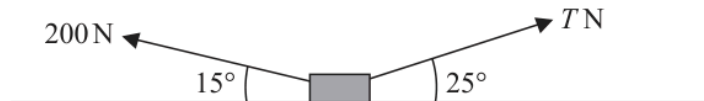


Figure 1

A parcel of mass 20 kg is at rest on a rough horizontal floor. The coefficient of friction between the parcel and the floor is 0.3

Two forces, both acting in the same vertical plane, of magnitudes 200 N and $T\text{ N}$ are applied to the parcel. The line of action of the 200 N force makes an angle of 15° with the horizontal and the line of action of the $T\text{ N}$ force makes an angle of 25° with the horizontal, as shown in Figure 1. The parcel is modelled as a particle P .

Find the smallest value of T for which P remains in equilibrium.

(9)

Worked Solution - Question 3

1. Resolve vertically

For vertical equilibrium, $R + 200 \sin 15^\circ + T \sin 25^\circ = 20g$.

2. Resolve horizontally

For the smallest T , friction is limiting and balances the remaining horizontal tendency. Horizontally, $200 \cos 15^\circ - T \cos 25^\circ - F = 0$.

3. Use limiting friction

$$F = 0.3R.$$

4. Eliminate R and F

From the vertical equation, $R = 20g - 200 \sin 15^\circ - T \sin 25^\circ$. Substitute into $F = 0.3R$ and then into the horizontal equation.

5. Solve for T

$200 \cos 15^\circ - T \cos 25^\circ - 0.3(20g - 200 \sin 15^\circ - T \sin 25^\circ) = 0$. Solving gives $T = 192.3 \dots \text{ N}$, so the smallest value is about **192 N**.

Final answer

$T = 192 \text{ N}$ approximately.

Question 4

Moments

4.

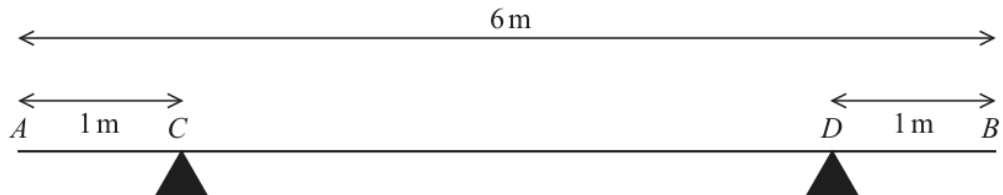


Figure 2

A metal girder AB has weight W newtons and length 6 m. The girder rests in a horizontal position on two supports C and D where $AC = DB = 1$ m, as shown in Figure 2.

When a force of magnitude 900 N is applied vertically upwards to the girder at A , the girder is about to tilt about D .

When a force of magnitude 1500 N is applied vertically upwards to the girder at B , the girder is about to tilt about C .

The girder is modelled as a non-uniform rod whose centre of mass is a distance x metres from A .

Find the value of x .

(6)

Worked Solution - Question 4

1. Set support positions

The girder has length **6** m, with C **1** m from A and D **1** m from B. So D is **5** m from A.

2. Use tilting about D

When the **900** N upward force is applied at A, the girder is about to tilt about D. Taking moments about D gives $900(5) = W(5 - x)$.

3. Use tilting about C

When the **1500** N upward force is applied at B, the girder is about to tilt about C. Taking moments about C gives $1500(5) = W(x - 1)$.

4. Solve the equations

The equations are $W(5 - x) = 4500$ and $W(x - 1) = 7500$. Add them:
 $W[(5 - x) + (x - 1)] = 12000$, so $4W = 12000$ and $W = 3000$.

5. Find x

Use $W(x - 1) = 7500$: $3000(x - 1) = 7500$, so $x - 1 = 2.5$ and $x = 3.5$ m.

Final answer

$x = 3.5$ m.

Question 5

Working with Vectors

5. A particle is acted upon by two forces $\mathbf{F}$ and $\mathbf{G}$. The force $\mathbf{F}$ has magnitude 8 N and acts in a direction with a bearing of 240° . The force $\mathbf{G}$ has magnitude 10 N and acts due South.

Given that $\mathbf{R} = \mathbf{F} + \mathbf{G}$, find

- (i) the magnitude of $\mathbf{R}$,
- (ii) the direction of $\mathbf{R}$, giving your answer as a bearing to the nearest degree.

(7)

Worked Solution - Question 5

1. Resolve $\mathbf{F}$ into components

A bearing of 240° is 60° west of south. Hence

$$\mathbf{F} = (-8 \sin 60^\circ)\mathbf{i} + (-8 \cos 60^\circ)\mathbf{j} = -4\sqrt{3}\mathbf{i} - 4\mathbf{j}.$$

2. Add $\mathbf{G}$

The force $\mathbf{G}$ is due south with magnitude 10, so $\mathbf{G} = -10\mathbf{j}$. Therefore

$$\mathbf{R} = \mathbf{F} + \mathbf{G} = -4\sqrt{3}\mathbf{i} - 14\mathbf{j}.$$

3. Find the magnitude

$$|\mathbf{R}| = \sqrt{(-4\sqrt{3})^2 + (-14)^2} = \sqrt{48 + 196} = \sqrt{244} \text{ N}.$$

4. Find the bearing angle

The resultant points south-west. The angle west of south satisfies

$$\tan \theta = \frac{4\sqrt{3}}{14} = \frac{2\sqrt{3}}{7}.$$

5. State the bearing

$\theta = 26.3^\circ$. Bearing $= 180^\circ + 26.3^\circ = 206.3^\circ$, so the bearing is 206° to the nearest degree.

Final answer

(i) $|\mathbf{R}| = \sqrt{244} \text{ N} \approx 15.6 \text{ N}$. (ii) bearing $= 206^\circ$.

Question 6

Working with Vectors

6. Two girls, Agatha and Brionie, are roller skating inside a large empty building. The girls are modelled as particles.

At time $t = 0$, Agatha is at the point with position vector $(11\mathbf{i} + 11\mathbf{j})\text{m}$ and Brionie is at the point with position vector $(7\mathbf{i} + 16\mathbf{j})\text{m}$. The position vectors are given relative to the door, O , and $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.

Agatha skates with constant velocity $(3\mathbf{i} - \mathbf{j})\text{ms}^{-1}$

Brionie skates with constant velocity $(4\mathbf{i} - 2\mathbf{j})\text{ms}^{-1}$

- (a) Find the position vector of Agatha at time t seconds. (2)

At time $t = 6$ seconds, Agatha passes through the point P .

- (b) Show that Brionie also passes through P and find the value of t when this occurs. (4)

At time t seconds, Agatha is at the point A and Brionie is at the point B .

- (c) Show that $\overrightarrow{AB} = [(t - 4)\mathbf{i} + (5 - t)\mathbf{j}]\text{m}$ (2)

- (d) Find the distance between the two girls when they are closest together. (4)

Worked Solution - Question 6

1. Write Agatha position

$$\mathbf{r}_A = (11\mathbf{i} + 11\mathbf{j}) + t(3\mathbf{i} - \mathbf{j}) = (11 + 3t)\mathbf{i} + (11 - t)\mathbf{j}.$$

2. Find point P

At $t = 6$, Agatha is at $(11 + 18)\mathbf{i} + (11 - 6)\mathbf{j} = 29\mathbf{i} + 5\mathbf{j}$.

3. Write Brionie position

$$\mathbf{r}_B = (7\mathbf{i} + 16\mathbf{j}) + t(4\mathbf{i} - 2\mathbf{j}) = (7 + 4t)\mathbf{i} + (16 - 2t)\mathbf{j}.$$

4. Show Brionie passes P

Set $7 + 4t = 29$ to get $t = 5.5$. Also $16 - 2t = 5$ gives $t = 5.5$. So Brionie also passes through P at $t = 5.5$ s.

5. Find AB vector

$$\overrightarrow{AB} = \mathbf{r}_B - \mathbf{r}_A = [7 + 4t - (11 + 3t)]\mathbf{i} + [16 - 2t - (11 - t)]\mathbf{j}.$$

6. Simplify AB

So $\overrightarrow{AB} = (t - 4)\mathbf{i} + (5 - t)\mathbf{j}$ m.

7. Minimise distance squared

$$AB^2 = (t - 4)^2 + (5 - t)^2 = 2(t - 4.5)^2 + 0.5.$$

8. State the minimum distance

The minimum occurs when $t = 4.5$, giving $AB^2 = 0.5$. Hence the minimum distance is $\sqrt{0.5} = 0.707 \dots$ m.

Final answer

(a) $\mathbf{r}_A = (11 + 3t)\mathbf{i} + (11 - t)\mathbf{j}$. (b) $t = 5.5$ s. (c) $\overrightarrow{AB} = (t - 4)\mathbf{i} + (5 - t)\mathbf{j}$. (d) minimum distance $= \sqrt{0.5}$ m ≈ 0.71 m.

Question 7

Kinematics Graphs

7. A helicopter is hovering at rest above horizontal ground at the point H . A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of 3.9 m s^{-2} for T seconds ($T < 6$), until his speed is reduced to $V \text{ m s}^{-1}$. He then moves with this constant speed $V \text{ m s}^{-1}$ until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves H and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

- (a) Find the speed of the parachutist at the instant when his parachute opens. (1)
- (b) Sketch a speed-time graph for the motion of the parachutist from the instant when he leaves H to the instant when he hits the ground. (2)
- (c) Find the value of T . (5)
- (d) Find, to the nearest metre, the height of the point H above the ground. (4)

Worked Solution - Question 7

1. Find speed when parachute opens

The parachutist falls freely from rest for 2.5 s, so $v = gt = 9.8(2.5) = 24.5 \text{ m s}^{-1}$.

2. Describe the graph

The speed-time graph starts at 0 , rises linearly to 24.5 at $t = 2.5$, then decreases linearly with smaller gradient for T seconds, then stays horizontal at speed V until $t = 20$.

3. Use the deceleration distance

During the deceleration phase, initial speed is 24.5 , acceleration is -3.9 , time is T , and distance is 73.75 .

4. Form the equation for T

$$73.75 = 24.5T + \frac{1}{2}(-3.9)T^2 = 24.5T - 1.95T^2.$$

5. Choose the valid T

Solving gives $T = 5$ or $T = 7.56 \dots$. Since $T < 6$, take $T = 5$.

6. Find V

$$V = 24.5 - 3.9(5) = 5 \text{ m s}^{-1}.$$

7. Find the total height

Height is the total area under the speed-time graph: first triangle

$$\frac{1}{2}(2.5)(24.5) = 30.625, \text{ deceleration distance } 73.75, \text{ and final rectangle } (20 - 2.5 - 5)(5) = 62.5.$$

8. Round the height

Total height = $30.625 + 73.75 + 62.5 = 166.875$ m, which rounds to **167** m.

Final answer

(a) 24.5 m s^{-1} . (b) Speed-time graph rises to **24.5** at $t = 2.5$, falls to V at $t = 2.5 + T$, then is horizontal to $t = 20$.

(c) $T = 5$. (d) $H = 167 \text{ m}$ to the nearest metre.

Question 8

Resolving Forces, Inclined Planes

8.

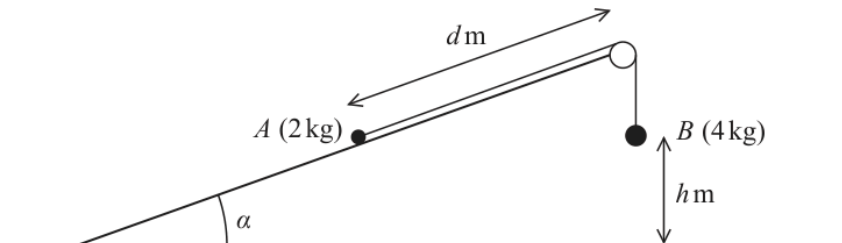


Figure 3

Two particles, A and B , have masses 2 kg and 4 kg respectively. The particles are connected by a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a rough plane. The plane is inclined to the horizontal ground at an angle α where $\tan \alpha = \frac{3}{4}$. The particle A is held at rest on the plane at a distance d metres from the pulley. The particle B hangs freely at rest, vertically below the pulley, at a distance h metres above the ground, as shown in Figure 3. The part of the string between A and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between A and the plane is $\frac{1}{4}$.

The system is released from rest with the string taut and B descends.

- (a) Find the tension in the string as B descends.

(9)

On hitting the ground, B immediately comes to rest.

Given that A comes to rest before reaching the pulley,

- (b) find, in terms of h , the range of possible values of d .

(7)

- (c) State one physical factor, other than air resistance, that could be taken into account to make the model described above more realistic.

(1)

(Total 17 marks)

Worked Solution - Question 8

Topic group

1. Use the trig ratios

Since $\tan \alpha = \frac{3}{4}$, $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$.

2. Find friction on A

For A, $R = 2g \cos \alpha = \frac{8g}{5}$. With coefficient $\frac{1}{4}$, friction is $F = \frac{1}{4}R = \frac{2g}{5}$.

3. Equation for A

A moves up the plane. Taking up the plane as positive: $T - 2g \sin \alpha - F = 2a$.

4. Simplify A's equation

$$T - 2g \cdot \frac{3}{5} - \frac{2g}{5} = 2a, \text{ so } T - \frac{8g}{5} = 2a.$$

5. Equation for B

B descends. Taking downward as positive: $4g - T = 4a$.

6. Solve for T

From A, $T = 2a + \frac{8g}{5}$. From B, $T = 4g - 4a$. Equating gives $6a = \frac{12g}{5}$, so $a = \frac{2g}{5}$. Then $T = 4g - 4\left(\frac{2g}{5}\right) = \frac{12g}{5}$ N.

7. Find speed when B reaches the ground

Before B hits the ground, both particles move distance h from rest with acceleration $\frac{2g}{5}$. Thus $v^2 = 2\left(\frac{2g}{5}\right)h = \frac{4gh}{5}$.

8. Find extra distance after B stops

After B hits the ground, A continues up the plane. The forces down the plane on A are $2g \sin \alpha + F = \frac{6g}{5} + \frac{2g}{5} = \frac{8g}{5}$, so its deceleration is $\frac{4g}{5}$.

9. Calculate extra distance

Let the extra distance be s . Then $0 = \frac{4gh}{5} - 2 \left(\frac{4g}{5} \right) s$, so $s = \frac{h}{2}$.

10. Find the range for d

A travels total distance $h + \frac{h}{2} = \frac{3h}{2}$ before stopping. Since it comes to rest before reaching the pulley, $d > \frac{3h}{2}$.

11. Name a more realistic factor

A more realistic model could include friction at the pulley.

Final answer

(a) $T = \frac{12g}{5} \text{ N} = 23.5 \text{ N}$. (b) $d > \frac{3h}{2}$. (c) A more realistic model could include friction at the pulley.